Chapter 12

Beyond Strings: In Search of M-Theory

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n his long search for a unified theory, Einstein reflected on whether "God could have made the Universe in a different way; that is, whether the necessity of logical simplicity leaves any freedom at all."1

With this remark, Einstein articulated the nascent form of a view that is currently shared by many physicists: If there is a final theory of nature, one of the most convincing arguments in support of its particular form would be that the theory couldn't be otherwise. The ultimate theory should take the form that it does because it is the unique explanatory framework capable of describing the universe without running up against any internal inconsistencies or logical absurdities. Such a theory would declare that things are the way they are because they have to be that way. Any and all variations, no matter how small, lead to a theory that—like the phrase "This sentence is a lie"—sows the seeds of its own destruction.

Establishing such inevitability in the structure of the universe would take us a long way toward coming to grips with some of the deepest questions of the ages. These questions emphasize the mystery surrounding who or what made the seemingly innumerable choices apparently required to design our universe. Inevitability answers these questions by erasing the options. Inevitability means that, in actuality, there are no choices. Inevitability declares that the universe could not have been different. As we will discuss in Chapter 14, nothing ensures that the universe is so tightly constructed. Nevertheless, the pursuit of such rigidity in the laws of nature lies at the heart of the unification program in modern physics.

By the late 1980s, it appeared to physicists that although string theory came close to providing a unique picture of the universe, it did not quite make the grade. There were two reasons for this. First, as briefly noted in Chapter 7, physicists found that there were actually five different versions of string theory. You may recall that they are called the Type I, Type IIA, Type IIB, Heterotic O(32) (Heterotic-O, for short), and Heterotic E8 × E8 (Heterotic-E, for short) theories. They all share many basic features—their vibrational patterns